Financial KPI Analysis for a Startup

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1. Introduction

Early-stage startups often lose track of whether they are spending money wisely. Founders and investors constantly ask the same questions: are we spending too much to get one customer? Is that customer even worth what we spent? And how much time do we have left before the money runs out? This project builds a simple financial model that answers exactly these questions using clear, easy-to-read numbers and charts.

2. Abstract

This project analyzes 24 months of a simulated early-stage startup's finances — revenue, expenses, customers, and marketing spend. From this, it calculates the two most important health numbers in startup finance: **CAC** (how much it costs to get one new customer) and **LTV** (how much that customer is worth over their whole time with the company). It also tracks how long customers typically stay (cohort retention) and how much cash runway the company has left. All of this comes together in a live, self-updating Excel dashboard that anyone — technical or not — can open and understand in seconds.

3. Tools Used

- **Python (Google Colab)** — to generate and calculate the financial data
- **Pandas & NumPy** — to build and process the numbers
- **Seaborn** — to build the cohort retention heatmap
- **Microsoft Excel** — to build the live KPI dashboard, formulas, conditional formatting, and charts

4. Steps Involved in Building the Project

- **Built 24 months of realistic startup data** — customers growing steadily, revenue following customer growth, and expenses that start high and slowly improve (a real early-stage spending pattern).
- **Calculated Burn Rate and Cash Runway** — how much money the company loses each month, and how many months of cash remain at that pace.
- **Calculated CAC (Customer Acquisition Cost)** — marketing spend divided by how many new customers joined that month.
- **Calculated LTV (Customer Lifetime Value)** — using each customer's average monthly spend and how long they typically stay before leaving (based on churn rate).
- **Calculated the LTV:CAC Ratio** — the single most important number in the project, showing whether the company earns more from a customer than it spent to get them.
- **Built a Cohort Retention model** — grouped customers by the month they joined and tracked what percentage of each group was still active over the following months, shown as a heatmap.
- **Built a live Excel dashboard** — four headline KPI numbers (with color-coded conditional formatting for at-a-glance health checks), three trend charts (Revenue vs Expenses, Customer Growth, LTV:CAC Ratio), and the cohort retention heatmap, all in one page.

5. Conclusion

By month 24, the startup reaches a monthly revenue of **Rs. 1,31,661.89** against a monthly burn rate of **Rs. 23,216.70**, leaving about **9.2 months** of cash runway at the current spending pace. The LTV:CAC ratio stands at **10.99** — well above the widely-used healthy benchmark of 3 — and stays consistently in the 8–14 range once the noisy first month (too few customers to be reliable) is excluded.

This tells a clear story: the company's unit economics are genuinely strong — it earns far more from each customer than it spends to acquire them. But a ratio this high can also mean the opposite problem: the startup may be **under-investing** in growth. Since customers are this valuable, spending more on marketing while there is still runway left could accelerate growth without hurting profitability. In short: the business model works — the next move should be to grow faster, not to cut costs.